MICROCONTROLLER · DATASHEET

ESP32-C3 Super Mini

board marking · USB-C · onboard PCB antenna

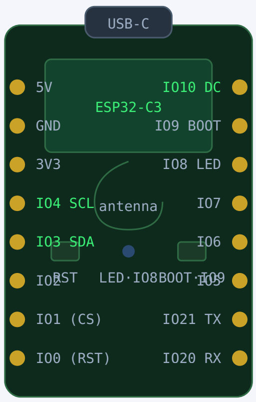
Wi-Fi + BLE SoC

RISC-V

single-core · 160 MHz

A thumb-sized development board built around Espressif's ESP32-C3 — a 32-bit RISC-V microcontroller with 2.4 GHz Wi-Fi and Bluetooth 5 Low Energy on a single die. In this project it is the whole computer: it joins your Wi-Fi, pulls live flight and weather data, and drives the round display over SPI. Powered and flashed through one USB-C port.

CPU RISC-V 32-bit @160 MHz	MEMORY 400 KB SRAM + 384 KB ROM
FLASH (MODULE) 4 MB typ.	RADIO Wi-Fi b/g/n + BLE 5
LOGIC LEVEL 3.3 V	POWER IN USB-C 5 V → 3V3 reg



representative pinout · yellow = header pad

Electrical & core specification

PROCESSOR & MEMORY	
Core	
Clock	
SRAM	
RTC SRAM	
ROM	
Flash	

WIRELESS	
Wi-Fi	
Bluetooth	
Antenna	
Security	

I/O & PERIPHERALS	
GPIO	
Serial buses	
ADC	
Timers/ PWM	
USB	
Sensor	

POWER & ENVIRONMENT	
Operating voltage	
Board input	
Wi-Fi TX peak	
Deep sleep	
Temp range	

3.3 V part. GPIOs are not 5 V tolerant. Power the display from the board's 3V3 pin, and never drive a 5 V signal into any IO pin.

Onboard controls & notes

FEATURE	DETAIL
	On GPIO9. Reused as the page button in this project. Held during power-up it forces the ROM bootloader — handy for flashing, annoying if pressed by accident (just re-plug).
	Hardware reset. BOOT + RST is the manual flashing handshake.
	Onboard LED on GPIO8 (also a strapping pin — leave floating at boot).
	GPIO2, GPIO8, GPIO9 influence boot mode; fine for normal use once running.

As wired in the MUC Desk Radar

Only the display SPI bus and the onboard button are used — the rest of the pins are free for future add-ons.

- GPIO4 → display SCL
- GPIO3 → display SDA
- GPIO10 → display DC
- GPIO1 → display CS
- GPIO0 → display RST
- 3V3 / GND → display power
- GPIO9 → page button (onboard BOOT)